

ENGINEERING TECHNOLOGY - SHORT NOTE

PART 08

Basic Electronic Technology

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1. Fundamentals of Semiconductors

Intrinsic Semiconductors

Intrinsic semiconductors are pure semiconductor materials without any significant dopant species present. A key characteristic of intrinsic semiconductors is that their **electrical conductivity increases as temperature increases** due to thermal excitation of charge carriers.

Extrinsic Semiconductors

Extrinsic semiconductors are created by adding a small, precise quantity of impurities—specifically elements from **Group III** or **Group V** of the Periodic Table—into an intrinsic semiconductor lattice. This intentional addition of impurities is called **doping**.

The added impurities are referred to as *dopants*. Depending on the type of impurity added, extrinsic semiconductors are categorized into two types:

- **N-Type Semiconductors**
- **P-Type Semiconductors**

2. Types of Extrinsic Semiconductors

N-TYPE SEMICONDUCTOR

Formed by doping an intrinsic semiconductor (e.g., Silicon) with a **Group V element** (such as Arsenic, Phosphorus, or Antimony).

- **Majority Carriers:** Electrons
- **Minority Carriers:** Holes
- **Dopant Ion:** Positive Donor Ions

Doping generates extra free electrons into the conduction band.

P-TYPE SEMICONDUCTOR

Formed by doping an intrinsic semiconductor (e.g., Silicon) with a **Group III element** (such as Boron, Aluminium, or Gallium).

- **Majority Carriers:** Holes
- **Minority Carriers:** Electrons
- **Dopant Ion:** Negative Acceptor Ions

Doping creates electron vacancies known as holes in the valence band.

3. The P-N Junction

When a P-type semiconductor and an N-type semiconductor are joined together, a **P-N Junction** is formed. Diffusion of majority carriers occurs across the boundary: free electrons from the N-region cross over to recombine with holes in the P-region.

This creates a region near the junction devoid of mobile charge carriers, containing only fixed positive donor ions on the N side and negative acceptor ions on the P side. This region is called the **Depletion Layer (or Depletion Region)**.



Figure 1.1: Structure of a P-N Junction and Depletion Layer

A P-N junction diode operates under two primary biasing conditions:

1. **Forward Bias Condition:** Positive terminal of DC source connected to P-type, negative to N-type. Reduces depletion width and allows current flow.
2. **Reverse Bias Condition:** Positive terminal connected to N-type, negative to P-type. Widens depletion layer and blocks current flow (except tiny leakage current).

4. Biasing & I-V Characteristics

Forward Bias Characteristics

In forward bias, current starts flowing rapidly once the applied voltage exceeds the **Threshold Voltage (Barrier Potential, V_B)**:

- **Silicon (Si) Diode:** Threshold Voltage $V_B \approx 0.7 \text{ V}$
- **Germanium (Ge) Diode:** Threshold Voltage $V_B \approx 0.3 \text{ V}$

In a practical circuit with a 6V supply and a Silicon diode:

Diode Voltage Drop $V_1 = 0.7 \text{ V}$

Remaining Load Voltage $V_2 = 6.0 \text{ V} - 0.7 \text{ V} = 5.3 \text{ V}$

Reverse Bias Characteristics

In reverse bias, the depletion layer widens. Only a negligible reverse saturation current flows until the voltage reaches the **Peak Inverse Voltage (PIV)** or Breakdown Voltage, where reverse breakdown occurs.

5. Types of Diodes and Their Applications

Diode Type	Primary Application / Function	Symbol Description
Rectifier Diode	Converts Alternating Current (AC) into Direct Current (DC).	Standard P-N Junction Triangle-Bar Symbol
Point Contact / Small Signal Diode	High-frequency wave detection and fast switching.	Fine wire contact inside glass envelope
Zener Diode	Voltage regulation in reverse breakdown mode.	Diode symbol with bent cathode bar ('Z' shape)
Light Emitting Diode (LED)	Emits light under forward bias (Indicators, lighting, displays).	Standard diode symbol with outward arrows
Photodiode	Detects light intensity (optical sensors, light meters).	Standard diode symbol with inward arrows

6. Rectification Principles & Circuits

Rectification: The process of converting Alternating Current (AC) into Direct Current (DC).

A. Half-Wave Rectification

Uses a single diode to allow only one half-cycle (positive) of the AC wave to pass, blocking the opposite half-cycle.

Key Formulas for Half-Wave Rectifier:

- Peak Voltage: $V_p = 1.414 \times V_{rms}$
- Average DC Output Voltage: $V_{dc} = V_o = \frac{V_p}{\pi} = 0.318 V_p$
- In terms of RMS Input Voltage: $V_{dc} = 0.318 \times 1.414 V_{rms} \approx 0.45 V_{rms}$

B. Full-Wave Rectification

Converts both positive and negative half-cycles of the AC input into pulsating DC output. Can be achieved using either a **Center-Tapped Transformer (2 Diodes)** or a **Bridge Rectifier (4 Diodes)**.

Key Formulas for Full-Wave Rectifier:

- Average DC Output Voltage: $V_{dc} = V_o = \frac{2V_p}{\pi} = 0.637 V_p$
- In terms of RMS Input Voltage: $V_{dc} = 0.637 \times 1.414 V_{rms} \approx 0.9 V_{rms}$

Filtering & Smoothing

The raw output of rectifiers is pulsating DC containing ripple voltage. A **smoothing capacitor** connected in parallel across the output load charges during voltage peaks and discharges through the load during troughs, producing smooth DC voltage.

Additional filtering can be achieved using an LC filter circuit featuring an **Inductor (L)** in series and a **Capacitor (C)** in parallel.

7. Practical Applications & Diode Testing

Special Applications of Rectifier Diodes

- **Circuit Protection:** Placed in series with DC power inputs to protect circuits against reverse polarity damage.
- **Automatic Polarity Correction:** Connecting a full-wave bridge diode arrangement before the load ensures the correct polarity (+) and (-) is always supplied to the load, regardless of input polarity connection.
- **Bulb Brightness / Dimmer Control:** Switching a 1N4007 diode into series with an AC lamp blocks one half-cycle, reducing power consumption and dimming the lamp.
- **Inductive Load Protection (Flyback Diode):** Connected across relays or electric fans to protect drive circuitry from inductive voltage spikes.

Testing Diodes using a Multimeter

To test a diode using a multimeter set to resistance (Ω) or diode test mode:

- **Forward Bias Connection (Red lead to Anode, Black lead to Cathode):** Shows **Low Resistance** (or forward voltage drop $\sim 0.6\text{--}0.7\text{V}$ for Si).
- **Reverse Bias Connection (Red lead to Cathode, Black lead to Anode):** Shows **High Resistance / Open Circuit (OL)**.

8. Zener Diodes & Voltage Regulation

A **Zener Diode** is specially designed to operate safely in the reverse breakdown region without damaging the junction. The constant voltage maintained in reverse breakdown is called the **Zener Voltage (V_Z)**.

Zener Voltage Regulator Design

When used as a shunt regulator with a series resistor R_S :

- Maximum Power Dissipation: $P_d = I_{Zmax} \times V_Z$
- Maximum Zener Current: $I_{Zmax} = \frac{P_d}{V_Z}$
- Voltage Drop across Series Resistor: $V_R = V_S - V_Z$
- Series Resistance Calculation: $R_S = \frac{V_S - V_Z}{I_S}$

Zener diodes can also be placed back-to-back in anti-series to stabilize or clip AC voltage signals at $\pm V_Z$.

9. Light Emitting Diodes & Voltage Regulator ICs

Light Emitting Diode (LED) Calculations

LEDs require a current-limiting series resistor R to restrict current I_S to safe operating levels:

- Total Supply Voltage: $V_S = V_R + V_D$
- Series Current: $I_S = \frac{V_S - V_D}{R}$
- Required Series Resistor Value: $R = \frac{V_S - V_D}{I_S}$

Integrated Voltage Regulator ICs

For adjustable and precise voltage regulation, standard linear voltage regulator ICs are used:

- **LM317:** Adjustable Positive Voltage Regulator IC
- **LM337:** Adjustable Negative Voltage Regulator IC
- **Dual Voltage Power Supply:** Combining LM317 and LM337 with a center-tapped transformer and bridge rectifier provides symmetrical positive (+V) and negative (-V) power rails relative to ground.